30.007 Term Paper

Awaludin, N. D, Chong, X. X, Choo, S. S. W, Koh, S. X. Y, Lee, B. Y. J, Lee, S. Y. X, Tan, K. K. L, Tan, M. P. W.

Abstract—Public transport navigation is one of the greatest challenges for individuals with visual impairment, particularly at transitional points such as bus stops. While existing solutions address aspects of the first, middle and last-mile navigation, there have been few solutions which target the confusion derived from the process of bus identification and boarding. Confusion experienced by individuals with visual impairment results from the variability in bus positioning upon arrival, and the difficulty in differentiating between buses when they arrive in close succession.

Our objective is to increase the independence of the visually impaired, specifically via enhancing navigation around bus stops. We developed an assistive navigation system comprising of 3 sub-systems. The first sub-system utilizes three ultra-wide bands (UWB) installed on the bus stop, which utilizes the process of trilateration to determine the bus's position relative to the bus stop. The second sub-system, an anchor mounted on the bus linked to an LED indicator, notifies the bus driver that a visually impaired individual intends to board the bus. The third sub-system, a wearable which automatically syncs to a bus stop in close proximity, announces the available bus services via audio feedback, allowing multiple selection of preferred bus services. Upon arrival of the preferred bus service, the wearable alerts the user and provides real-time directional guidance, enabling them to accurately locate the bus entrance.

Prototype testing suggested that the size, weight, battery duration and audio feedback of the system were suitable for daily usage. However, user feedback suggested that implementation of audio feedback dependent on proximity to bus entrance, may improve the system's intuitiveness. Acknowledging areas for improvement is key in developing a solution which effectively addresses the needs of individuals with visual impairment. Although navigational obstacles are a common occurrence for individuals with visual impairment, it should not define their mobility. Thus, our system aims to increase transportation accessibility by reducing the confusion experienced by the visually impaired.

INTRODUCTION

Individuals of any age or background are susceptible to vision impairment. The common causes of visions impairment would be congenital conditions, accidents or illness. In Singapore, over 40,000 people are living with blindness or forms of vision impairment, a number which is projected to rise sharply as the population ages [1]. It was crucial to obtain ground feedback, discussions with Singapore Association of the Visually Handicapped (SAVH) and interviews with visually impaired individuals, such as Lyn, Head of Accessibility & Honorary Secretary, were conducted. These interactions surfaced up unique challenges and perspectives from an organizational and individual level, with regards to the use of public transport, in particular utilizing buses as a mode of transport.

Through our interactions with SAVH and individuals with visual impairments, a literature review of the state of the art was conducted, to understand the methodologies of pre-existing solutions, in hopes of identifying gaps present. Numerous assistive navigational solutions have been locally and internationally developed:

- All Aboard – Mobile based application which utilizes computer vision to assist individuals with visual impairment in identifying their bus stop.
- NaviLens – Quick Response (QR) identification system, reliant on line of sight to provide wayfinding and navigational assistance.
- Light Guide (Tactile Compass) – A handheld device which provides tactile feedback via visual imaging provided by a camera attached to the user.
- MAVIS – An integrated system which encompasses a bus timing application to allow for selection of a bus number to the control of audio announcements for boarding and alighting.
- Waymap SG – Navigational mobile based application which utilizes existing infrastructure blueprints along with LiDAR to provide navigation guidance.

Existing solutions employ a variety of technologies such as QR markers, Global Positioning System (GPS), tactile feedback, auditory feedback, to assist individuals with visual impairment in route planning and location positioning. These solutions are effective in addressing the needs of the visually impaired in relation to first, middle and last mile navigation of public transportation.

Despite various technologies present which provide navigational assistance for the visually impaired, they do not address issues experienced at transition points. The main issues experienced are:

1. Variability in bus positioning at a bus stop after coming to a stop.
2. Difficulty in identifying the correct bus entrance when buses arrive in close succession

Challenges that individuals with visual impairment experience while utilizing existing products, are categorized into primary, secondary and tertiary, ranging from highest to lowest of importance respectively.

Primary: Product's dependency on infrastructure for usage, lack of accurate, real-time directional guidance at bus stops.
Inability to differentiate between bus stops, in the presence of opposing bus stops.

Secondary: Reliance on public to verify preferred bus services, visual confirmations to ascertain relative positioning at bus stop. Preference of auditory or tactile feedback for directional feedback

Tertiary: Method of arrival to transition points, bus stops. As pre-existing solutions effectively address navigational challenges in the first to last mile segment of commuting.

Through the identification of gaps addressed in pre-existing solutions, the hypothesis is an assistive bus stop navigational system which offers real time positioning and guidance, possesses the ability to improve independence and reduce confusion experienced

by individuals with visual impairment. The proposed solution integrates the trilateration positioning from ultra-wide bands (UWB) with a wearable that provides auditory feedback for navigational guidance. The solution primarily targets the last steps in navigation, from bus stop to appropriate bus entrance. The solution tackles an untouched area, where no existing solution offers reliable, intuitive, real-time navigational aid. The solution's specific focus about bus stops aims to improve independence of individuals with visual impairment, ensuring that urban transportation systems will be inclusive and accessible, regardless of disabilities.

I. Proposed Approach

A. System Requirements

The bus stop guidance system for the blind was developed to improve accessibility, confidence and independence for visually impaired users when boarding buses. Success is defined by meeting specific, objective and measurable parameters that evaluate user experience, implementation feasibility and technical performance. A series of trials in a simulated bus stop environment (with mock-up shelter, and incoming bicycle as the bus) were conducted to measure these parameters. The table below summarizes the system requirements.

Table I: Method of product assessment

Category	Sub-Criterion	Unit of Measurement	Target for Success and Result
User-testing	Ease of use	Likert Scale (1-5)	≥4.0 average (achieved)
	Intuitiveness	Time taken (minutes). To utilize devices independent	≤ 5 minutes (did not achieve)
Implementation	Portability	Weight (grams) Dimensions (centimeters)	Weight ≤ 250 grams Max. Dimension: 10cm (Length) x 7 cm (Weight) x 5 cm (Thickness)
	Cost	SGD ($)	Attachment ≤ $100 (achieved)
Accuracy	System latency	Time delay (seconds)	≤ 10 seconds
	Position accuracy	RMSE (meters)	≤ 10 meters (achieved)

B. Concept Selection

Two approaches were considered, mainly Computer Vision (CV) Detection and Ultra-Wide Band (UWB) Positioning.

CV offered the advantage of identifying bus numbers and entrance without modifying buses and could be deployed widely. However, its performance is sensitive to lighting, weather and visual clutter, and it requires significant processing power and large training datasets.

UWB positioning provided high accuracy (10–30 cm) and was unaffected by lighting or visual obstructions, making it reliable in varied environments. Its drawbacks include potential signal attenuation by metal, water, or human bodies, as well as higher power consumption and the need for anchor installation.

The final concept selected was UWB trilateration, as it offers consistent accuracy and robustness in outdoor conditions, and integrates well with the navigation cane to provide real-time vibration and audio feedback to the user.

C. Concept Description

The proposed system can be simplified to four separate sub-systems: The user's wearable device, the central bus hub, the on-board bus module, and the bus positioning system. Undergirding these sub-systems is a wireless communication system that facilitates the transmission of data and signals to relevant sub-systems.

Undoubtedly the greatest focus was placed on the user's module as it would be the device that visually impaired users directly interface with. However, developing the proposed concept into a functional product requires a more comprehensive system in conjunction with the device in the user's hand.

The first input of the system begins with the user. Upon powering up the device, users select the bus stop of their choice. Navigating through a list of clearly defined audio questions and cues, users can input their choice of buses and send their final selection to the central bus hub. (Figure 1.1)

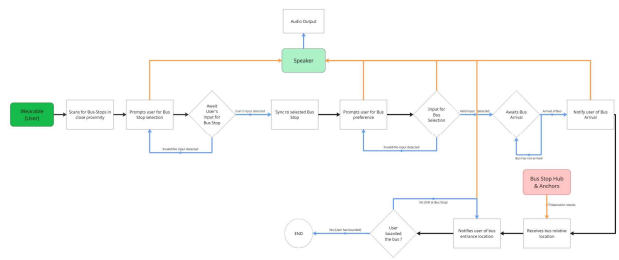

Figure 1.1. User device feedback control system.

Next, the bus hub receives the information sent by the user device and begins transmitting wireless messages to all approaching buses, informing these buses to stop if they are one of the buses in the list of user selected buses.

The main functionality of the on-board bus module is to receive the information transmitted by the central bus hub and parse through the selected bus list to see if the said bus is inside. If so, an indicator light turns on to notify the bus driver that a visually impaired user is waiting to board his bus at the oncoming bus stop. At the same time, an acknowledgement message is sent back to the central bus hub and rerouted to the user device to inform the user that his/her bus has arrived.

In addition to informing users that their selected buses have arrived, the bus positioning system also obtains information about bus positions with respect to the bus stop. This positional information is paired with the bus arrival data and presented to users through auditory feedback, giving users a clearer idea of what bus has arrived and where it is located at the bus stop.

To facilitate the effective transmission of data and information, we needed to develop a robust wireless communication network. As we were using ESP32's as the main microcontrollers for our devices, two options were presented to us: ESP classic BLE or ESPNOW communication protocol.

Initially we experimented with the ESP classic BLE for bus identification. The results seemed promising with the system being able to function reliably, achieving an average detection distance of 60 meters along an open road.

However, the ESP classic BLE's came with significant drawbacks as well. One of the main disadvantages was that that the maximum transmittable payload data was only 31 bytes. For early detection and bus identification with a single user, this payload size was sufficient. However, with the need to transmit more information such as available buses, selected buses, and arrived buses to not just one but multiple users, it was very likely we would exceed the 31 bytes limit. Moreover, because many other devices run on classic BLE, we found that the bus stop module was often loaded with the scan results of many unrelated BLE devices in the vicinity. Given the higher noise levels, the ESP BLE module must refresh scans at a significantly faster rate to scan through all BLE devices in the vicinity to find users' selected buses, making the system less energy and time efficient.

For these reasons we decided to pivot and use ESPNOW as our primary wireless communication channel. Unlike the ESP classic BLE, ESPNOW allows for payload data of up to 250 bytes. It can establish reliable connections between ESPNOW devices over distances exceeding 100m of open road (based on field tests). Additionally, as ESPNOW utilizes the onboard ESP Wi-Fi chip for communication, only other ESPNOW devices can send and receive data in the established channel, effectively minimizing the amount of noise that the ESPNOW devices must filter through.

In many applications of ESPNOW, connections between devices are predefined with the MAC addresses of each device. However, in our specific application, doing so would not be practical given the sheer number of MAC addresses that each device would have to be configured to (based on the number of buses, bus stops, and users). Instead, to allow for dynamic connections between devices, we utilized the universal broadcast MAC Address: FF:FF:FF:FF:FF:FF.

With each user being able to transmit and receive data from every other user, there was a need to filter through the received information so that devices would only act when they received new data that was relevant to them. This was achieved by including a filtering mechanism in the callback query handler section of the ESPNOW code (Figure 1.2).

```
// callback when data is received
void OnDataRecv(const esp_now_recv_info_t *recv_info, const uint8_t *incomingData, in

  memcpy(&tempcheck, incomingData, sizeof(tempcheck));

  if (tempcheck.bus_stop_id == 0) { //filters out all non bus stop devices.
    return;
  }
  else if (state != 1) { //not in choose_bus state
    if (tempcheck.bus_stop_id != my_bus_stop) { //check if bus that user selected.
      return;
    }
  }

  memcpy(&incomingReadings, incomingData, sizeof(incomingReadings)); //take a snapsho
```

Figure 1.2. Callback query broadcast filtering.

Moreover, to reduce the need to parse through the entire data payload every time a new payload was received, we utilized the structured message functionality of ESPNOW to format payload data. Additionally, we also optimized the system memory loading by opting to transmit signed integers (4 bytes of memory per signed integer) instead of string data types (24 bytes of memory per string) (Figure 1.3).

```
typedef struct struct_message_1
{
  String bus_stop_id;
  String bus_stop_buses[3];
  String arrived_buses[10];
  String selected_buses[10];
} struct_message_1;
```

```
typedef struct struct_message_1
{
  String user_name = "sam";
  int bus_stop_id = 0;
  int bus_stop_buses[bus_stop_buses_size];
  int arrived_buses[arrived_buses_size];
  int selected_buses[selected_buses_size];
  int bus_service = 0;
} struct_message_1;
```

Figure 1.3. Broadcast message structure and data types.

II. Modelling

Since the system consists of both hardware prototyping and software IOT, calculations were required to choose the appropriate materials. For the hardware side, battery sizing and comparison of materials were done, using calculations on capacity requirement and materials strength comparisons. For IOT, analysis of antenna radiation patterns and implementation of a Kalman filter was used for performance predictions, along with trilateration for bus positioning.

A. Component Selection

One of the considerations was for the device to have a long battery life, requiring less charges between uses, and having a long-life span. The battery was thus sized to our context, for up to 18 uses before requiring charging. Taking the operating current of the ESP32 and speaker, along with the operation time per use, we calculated the capacity required for the system to be 118.7mAh. Sizing for 18 uses, accounting for a depth of discharge of 80%, to ensure battery health, and a safety factor of 1.1, our final capacity requirement was 2937mAh. We then calculated the peak current draw of the system, with 20% tolerance, to be 1264.8mA. Our battery requirements were as such: Capacity ≥ 3000mAh, with a discharge rate ≥ 0.5. Hence, we chose a lipo-battery that meets both capacity and discharge rate requirements. A secondary consideration for battery was the size and weight of the battery as our device aims to be small and lightweight.

B. Performance Predictions

To continuously track the position of buses without visual detection, trilateration was accomplished using UWB antennas. UWB antennas can perform localization using these protocols: Time Difference of Arrival (TDOA) and Two-Way Ranging (TWR).

In TDOA mode, each tag representing the bus will periodically transmit a pulse which would be detected by multiple anchors placed at bus stops. As each anchor is placed at different distances, they will receive the tag's pulse at slightly different times. Based on these time differences, the server can calculate the tag's position. In TWR mode, the anchor will send an ultra-wideband pulse signal and receive a reply from the tag on the bus. The flight time can be used to calculate the distance between the anchor and the tag.

The method for localization was chosen after considering the benefits of both protocols. Although TDOA requires lower power consumption than TWR, it offers relatively poor positioning accuracy outside of the anchor area. TWR does not require a master tag and will be able to scale in both number of tags and anchors, making it suitable for the application of tracking buses at bus stops. Therefore, the method for localization in the application of bus tracking was TWR.

ESP32 UWB DW3000 modules were utilized for the localization of buses, and the radiation pattern of these modules would have to be accurately modelled out to aid in our testing and determination of optimal anchor and tag placements. The radiation pattern was first tested with a single UWB module and verified that it matched the datasheet, as seen in Fig 3.1, allowing us to use the datasheet pattern for our mathematical modelling.

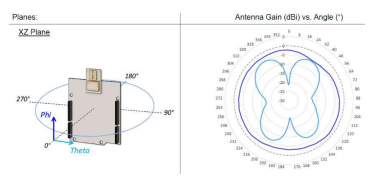

Figure 2.1. Datasheet radiation pattern

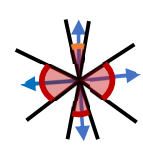

Figure 2.2. Actual radiation pattern

From the radiation pattern provided, we determined the angles of highest magnitude of dBi (blue arrows), along with the angles at our acceptable signal value of -10dBi (black lines), as seen in Fig 2.2.

Since the radiation pattern is not omni-directional, it was crucial to geometrically work out the angles to ensure that both the anchors and the tags could resonate. Given the specifications for bus stops set by Land Transport Authority (LTA), the design goals were as follows: At least 2 distinct positions for a bus stop of 27m, an error margin of at least ±2m for each position. Given the boundary constraints of the bus stop width of 3m, and bus length of 12m, an assumption that the optimal bus positions will be 14m apart was made, since the buses would have some gap, and the max width of the system would be 4m, since the bus may stop slightly further from the edge of the bus stop. Due to the nature of the radiation pattern seen in Fig 2.2, there are blind spots enveloping each acceptable radiation range, whilst the strongest signals are about 90° apart. As such, the only geometry that will allow a single anchor to potentially radiate to 2 bus positions 14m apart would be a triangle. Hence, our group analysed isosceles and scalene triangles. Isosceles triangles would give the most balanced radiation pattern as seen in Fig 2.3. However, the only position that the isosceles triangle can fit is at y = 4m, due to the tight spread of the signal attenuation. As such, scalene triangles were considered next. As seen in Fig 2.4, we chose the tighter grouping to be facing towards the longest side of the scalene triangle, as a change in distance along the bus positioning line would lead to a less than proportional increase in the angle required, keeping it within the effective radiation of the UWB anchor. Since the anchors need to be arranged with at least 2 with different y values, and at least 2 with different x values, the final positions were determined, as seen in Fig 2.5, taking the front bus position to be origin. The final values are tabulated as such:

Table II: Coordinates of anchor points

Anchor	x-coordinate	y-coordinate	Angle
A1	-7	4	120°
A2	-12	3	60°
A3	-12	4	60°

Since the radiation patterns of both bus stop anchors and bus tag antennas must match, the bus tag antenna placements would then have to be determined as well. To do so, we drew out the radiation patterns given by the 3 anchors to the 2 optimal bus positions, B1 and B2, with a ±2m tolerance each. We then orientated the bus tag's antenna such that it would align with the radiation lines as seen in Fig 2.5. Doing so enabled us to determine that the optimal angle for the bus tag is 65°.

The trilateration of 3 anchors equations was derived using the positions of each anchor shown below, with anchors labelled as A1, A2, A3, with their respective x_i, y_i, r_i.

$$\hat{x} = \frac{(r_1^2-r_2^2-x_1^2+x_2^2-y_1^2+y_2^2)[2(y_3-y_2)]-[2(y_2-y_1](r_2^2-r_3^2-x_2^2+x_3^2-y_2^2+y_3^2)}{4[(x_2-x_1)(y_3-y_2)]-4[(x_3-x_2)(y_2-y_1)]} \quad (1)$$

$$\hat{y} = \frac{[2(x_2-x_1)](r_2^2-r_3^2-x_2^2+x_3^2-y_2^2+y_3^2)-(r_1^2-r_2^2-x_1^2+x_2^2-y_1^2+y_2^2)[2(x_3-x_2)]}{4[(x_2-x_1)(y_3-y_2)]-4[(x_3-x_2)(y_2-y_1)]} \quad (2)$$

Both x and y coordinates are determined based on the set anchor positions, and from the equations, a greater difference in x and y coordinates will lead to a more robust prediction as changes arising from environmental inaccuracies will be smaller as the difference in x and y coordinates increases. The r measurements are done by the UWB module and are not controllable. However, by analyzing the equations, the r_2 term is repeated 4 times as compared to r_1 and r_3 terms. As such, it can be deduced that the r_2 term places a bigger role in affecting the $\hat{x}$ and $\hat{y}$ values. As such, we made sure to position anchor 2 in the best position of the 3 anchors, making our system more robust.

Due to the highly directional antenna radiation pattern of the UWB antenna, a Kalman filter was implemented to support the trilateration of buses. The Kalman filter is an estimation algorithm that can estimate the state of the system, even when measurements are imprecise and uncertain, and is commonly used in tracking and data prediction systems.

The true state of the tracked object can be represented as x_t and the measured state of the object can be represented as z_t as seen in Fig 3.6. x_t of the object can be expressed using its previous state together with a disturbance in the system, process variance (Q_t). z_t can be expressed as the observed true state using sensors or localization, as well as the measurement variance (R_t). The predicted coordinates of each bus can be adjusted using Q_t and R_t to account for inaccuracies and measurement errors. Therefore, a 2D Kalman filter for the x and y values was used to predict the x and y value of the moving buses.

$$\begin{cases} x_t = F_t x_{t-1} + w_t \\ z_t = H_t x_t + v_t \end{cases},$$

$$\text{where} \begin{cases} x_t : \text{true state at time } t \text{ (a vector of real numbers)} \\ z_t : \text{measurement at time } t \text{ (or observation)} \\ F_t : \text{state transition model applied to } x_{t-1} \\ H_t : \text{observation model} \\ Q_t : \text{process noise (co) variance, } w_t \sim \mathcal{N}(0, Q_t) \\ R_t : \text{measurement noise (co) variance, } v_t \sim \mathcal{N}(0, R_t) \end{cases}$$

Figure 2.6. Kalman Filter Equation

III. Results

A. Prototypes

Our prototype development focused on creating a functional wearable device that meets ergonomic and technical requirements. The features are summarized as such:

- Simple two button interface for skipping and confirming options
- Anti-slip textured grip along the sides of the device for secure handling
- Audio feedback from a cavity speaker
- Attachment point for lanyard or strap connection

The button design has a raised profile of 10mm to provide tactile landmarks and are optimally placed at the top corner of the device for easy one-handed usage. Raised dots design was chosen due to its enhanced tactile feedback, as compared to alternative designs such as ribbed or knurled texture patterns. The intentional use of a cavity speaker is due to its space-efficient profile that accommodates the compact form factor of the device, while also maintaining sufficient audio output for outdoor environments. Moreover, the placement of the speaker in the center of the device ensures that users will not unknowingly cover the audio output. Lastly, the attachment point ensures that users can secure the device to prevent accidental drops during use.

The initial ergonomic design was symmetric to achieve a clean form and straightforward usability. However, it was modified to an asymmetric L-shaped configuration due to circuitry constraints which required the UWB module placement to be off-centered at the bottom of the device. The thickness of the device had to increase such that electrical components, such as the Lipo battery, speaker module and DF player, are not overlapping to prevent UWB signal interference from metallic parts. However, this modification is advantageous for the intended single-handed usage, specifically optimized for left-hand use while the right hand holds the walking cane. The resulting form factor naturally aligns with hand grip patterns and provides intuitive button access without compromising

navigation aid usage. In terms of material choice, Acrylonitrile Styrene Acrylate (ASA) was selected as the casing material. ASA provides 42-45 MPa tensile strength and 19-20 kJ/m² impact resistance, combined with excellent weather resistance and low water absorption (1%), ensuring reliable outdoor performance and structural integrity during daily use [2].

The following simulations were conducted:

Static stress loading was conducted to analyze deformation patterns under different conditions. Three scenarios were tested: (i) a force of 1000 N applied on the top surface, (ii) a force of 1000 N applied on the side surface, and (iii) a pulling force of 50 N applied at the attachment point where the lanyard would be secured. The 1000 N load was chosen to simulate the weight of an individual of approximately 100 kg sitting on the device, while the pulling force represents tension from the lanyard. The forces used are an overestimation, providing a conservative safety margin, and all simulations were performed using the material properties of ASA.

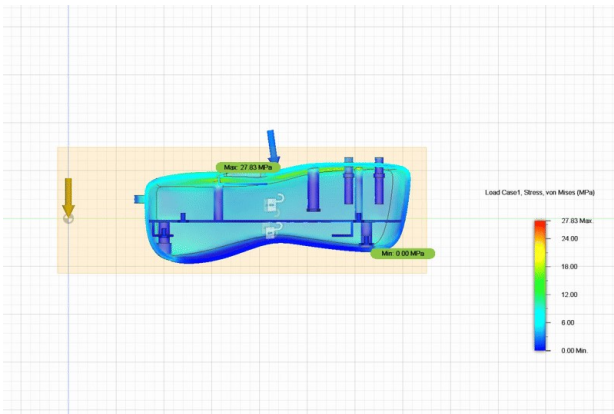

Figure 3.1. Static Stress simulation of the top surface

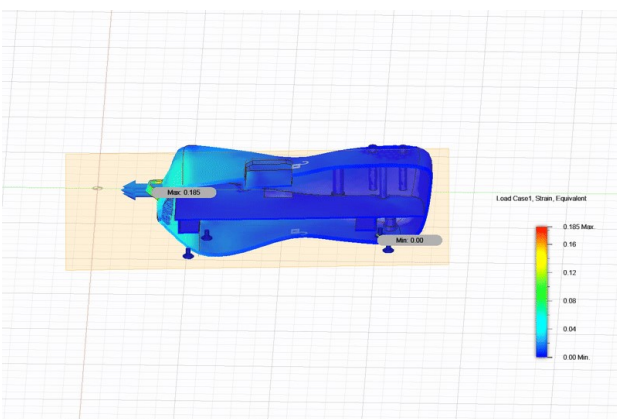

Figure 3.2. Static Strain simulation of the attachment point

Among the static load cases, the highest stress was observed when the top surface was loaded, reaching a maximum of 27.83 MPa. For the pulling force scenario, the maximum strain recorded was 0.185.

A dynamic event simulation was conducted to model real-world conditions, including accidental drops in various orientations. An average impact velocity, calculated from a drop height of 1.25 m, was used as the input parameter for these simulations.

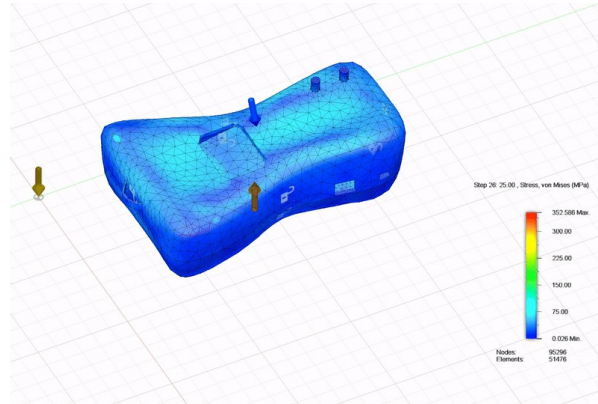

Figure 3.3. Dynamic Stress simulation of the top surface

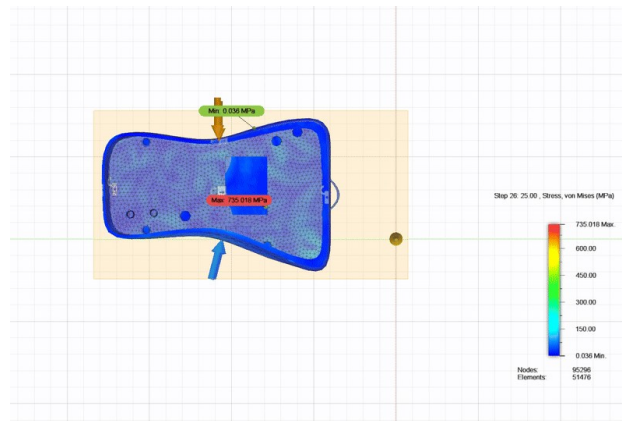

Figure 3.4. Dynamic Stress simulation of the side surface

The maximum stress recorded was 352.586 MPa when the impact force was applied to the top surface of the device, and 735.018 MPa when applied to the side surface.

The results from the static and dynamic simulations demonstrate that the device is structurally capable of withstanding conservative loading conditions and real-world impact scenarios, being within the acceptable limits for ASA material performance. The dynamic event simulations confirm that the device maintains its structural integrity when subjected to drops from a height of 1.25 m across multiple orientations. These findings suggest that the current design provides sufficient durability and reliability for practical use.

B. Experiments

The calculated values for trilateration gave us the desired results at the 2 optimal bus positions, even with ±2m tolerance. However, a more accurate model would involve the bus having an initial velocity. Furthermore, as seen in Fig 2.3 and Fig 2.4, there are clear blind spots that the UWBs will fail to communicate with each other. Due to this sudden blindness, the UWB modules would take significant time to update when the bus moves from the back position to the front position. As such, there was a need to implement the Kalman filter to smooth out the predicted results by taking reference from the past results, filling in the gaps of the blind spot. However, due to the Kalman filter, when the bus stops at the forward position, despite the anchors being able to read the bus at that position, the Kalman filter will still cause predictions to slightly overshoot before stabilizing at the correct position. As such, there is a need to adjust the Kalman filter parameters to minimize overshoot, whilst ensuring that the response time of the system meets our requirements, that is about 20s, as the bus typically stops for about 30s, and the user requires time to react and move to the bus. The optimal parameters were experimentally determined by changing the measurement variance R_t and process variance Q_t with different possible combinations and understanding the effects on localization.

Based on the initial parameter values of measurement and process variance, a hypothesis was formed. Since the sensor values varied significantly when not in the optimal direction, the measurement variance R_t should be increased to account for the inaccuracies of the anchor measurement device. The increase in measurement variance R_t would lead to a lower reliance on the measured values and depend more on the prediction model. Secondly, the process variance Q_t should also be increased. Given that the initial prediction model predicted that the bus continued to travel forward even when the tracked object has stopped, a secondary hypothesis that the model does not sufficiently account for external disturbances within the system was made.

Testing to prove the hypotheses was conducted separately. Keeping Q_t constant at the initial value of 10^{-4}, R_t was varied from 150 to 230 in increments of 20. Test conditions included modelling a bus stop dimension of 27m, front and rear bus positions, and a bicycle with a speedometer to repeatedly conduct the test at a constant speed. However, since the speed of the bicycle would ultimately depend on the rider, there would invariably be random errors with the test. To minimize random errors, only a single rider was used in addition to the speedometer.

The results from the test could be seen in Fig 3.5 below. Using the sum of root-mean-square error of multiple bus positions within the bus stop as a measurement of accuracy, the relationship that the accuracy of the bus prediction would improve with an increase in measurement variance was justified. In addition to accuracy, the latency of bus detection was also critical to the system's functionality. The time taken for bus detection with varying R_t was measured and averaged. Results proved that the detection time did not correspond to R_t, given that the difference in detection time between the fastest and slowest was only 9.04%, within 1 second. Since human detection error would be within the range of 1 second, the result would be insufficient to prove any relationship between R_t and latency.

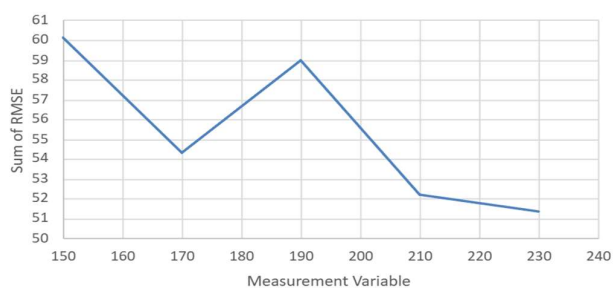

Figure 3.5. Sum of Root-mean-square against R_t

The test varying Q_t constant was conducted with the best R_t value of 230 for all experiments. However, the results are inconclusive. Since detection time for repeated experiments varied by up to 50% of the value, it would not be able to provide a conclusion. As such, the final parameters were determined to be 230 and 10^{-4} for R_t and Q_t respectively.

IV. Discussion

Based on conducted field tests, we found that it is possible to build a bus identification and positional tracking system for the visually impaired. The bus identification system functions reliably with minimal latency across 100m of straight open road using the ESPNOW wireless communication network. As for the bus positional tracking system, we demonstrated its ability to determine which of the three locales (front, middle, back) the bus is located at, with an accuracy of ±2m. This was achieved using the UWB ranging capabilities of the DWM3000 modules. On the user device side, we found from our user interviews that the device felt slightly bulky, but most of the users praised its overall curved ergonomic form factor. Nevertheless, they mentioned that this was not their primary consideration, citing that they were more concerned about the robustness of the functioning system. Moreover, the system also needs to account for unforeseen scenarios, given the dynamic nature of bus commuting. Some examples that we received from user feedback include: The arrival of multiple selected buses, the arrival of two identical bus services, interference with the buses and bus stops on the opposite side of the street, incorrect selected bus inputs, etc.

V. Conclusions

Development of an assistive, bus stop navigational system is crucial. As it aids in spearheading innovation in the field, to address the lack of navigational assistance and confusion experienced by individuals with visual impairments. Consolidation of feedback from user testing and system experimentations suggests that the integration of UWB modules, communication protocols, ESPNOW, and a wearable with auditory and tactile inputs possess the potential to reduce confusion and increase navigational awareness, through real-time bus notification and directional feedback. Beyond the solution's specific application, the solution aims to ensure that, at every phase of a user's journey, public transportation remains accessible and inclusive for all, regardless of any disabilities or impairments. Despite the validation of the system's effectiveness, further development is necessary to ensure the improved robustness and adaptability of the system. Areas of improvement are miniaturization of the wearable, integration of omni-directional antennas in UWB modules, and incorporation with existing official transportation infrastructure. Further exploration of areas for improvement would ensure the solution achieves its core function of providing navigational assistance about bus stops. Through the solution, it may bolster the advancement of relevant technologies in line with assistive navigation, fostering an environment where universal inclusion and accessibility for individuals with disabilities is present.

Appendix

Appendixes should appear before the acknowledgment. You can include here your Gantt Chart, Budget details, FMEA, etc.

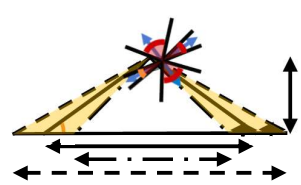

Figure 2.3: Scalene Triangle Configuration

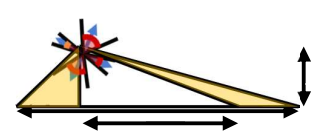

Figure 22.4: Isosceles Triangle configuration

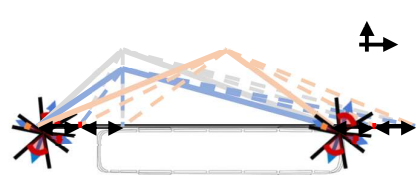

Figure 2.5. Variation of Anchor positioning

Table III: Failure Mode & Effects Analysis

SN	Function	Failure Mode	Effect	Causes	S	O	D	RPN

1	Tactile feedback	Failure to be in the correct position	Failure of system to give approximate location of bus	Repeated rough usage	8	3	10	240
2	Bus identification	Failure to detect bus	Failure to notify user that the bus has arrived, and guide user to bus	Electro-magnetic interference	8	3	10	240
3	Message transmission and reception	Failure to give quick response to user	Failure of system to notify and guide user within 10s	Latency of devices	7	4	4	112
4	Trilateration to correct bus	Trilateration error margin exceeds length of one bus	Failure to guide user to the correct bus	Large margin of error for UWB trilateration	8	2	5	80
5	Speaker	Failure to be heard	Failure to notify user of bus arrival	Crowded and noisy bus stops	7	3	4	84
6	User connection to correct bus stop	Failure to connect to correct bus stop	Incorrect bus stop connection leads to wrong guidance to user	UWB ranging, identifies opposite bus stop instead	8	2	3	48

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